Bayes Filter — Particle Filter and Monte Carlo Localization

Autonomous Navigation of Mobile Robots

Anna Semakova
a.semakova@gmail.com

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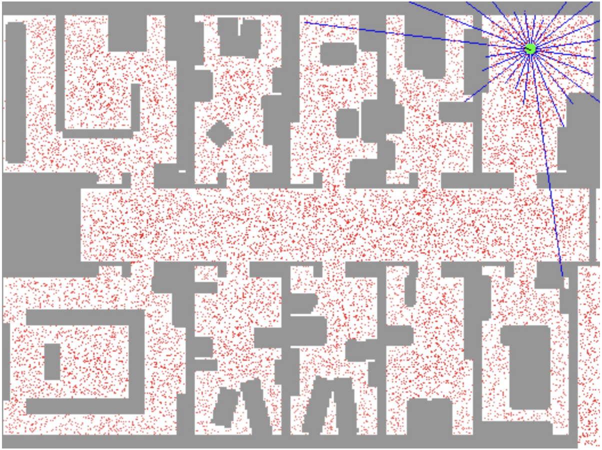
Motivation

- Recall: Discrete filter
  - Discretize the continuous state space
  - High memory complexity
  - Fixed resolution (does not adapt to the belief)
- Particle filters are a way to efficiently represent non−Gaussian distribution
- Basic principle
  - Set of state hypotheses (“particles”)
  - Survival−of−the−fittest

Sample−based Localization

Sonar



Mathematical Description

- Set of weighted samples

$$S = \{< s^{[i]}, w^{[i]} > \mid i = 1, \ldots, N\}$$

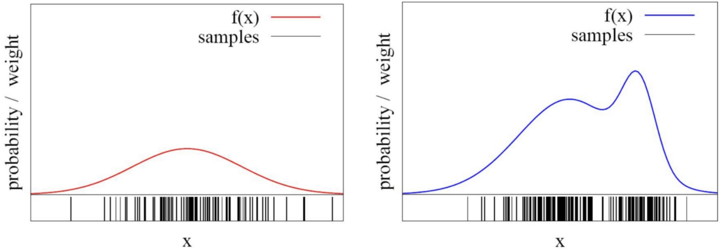
$s^{[i]}$ – state hypothesis
$w^{[i]}$ – importance weight

- The samples represent the posterior

$$p(x) = \sum_{i=1}^{N} w_i \cdot \delta_s[i](x)$$

Function Approximation

- Particle sets can be used to approximate functions



- The more particles fall into an interval, the higher the probability of that interval
- How to draw samples from a function/distribution?

Bayes filter with particle sets

- Measurement update

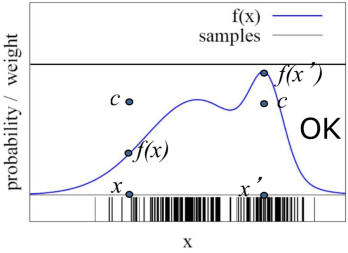
$$\begin{aligned} bel(x) &\leftarrow p(z \mid x)\overline{bel}(x) \\ &= p(z \mid x) \sum_i w_i \, \delta_{s^{[i]}}(x) = \sum_i p(z \mid s^{[i]}) w_i \, \delta_{s^{[i]}}(x) \end{aligned}$$

- Motion update

$$\begin{aligned} \overline{bel}(x) &\leftarrow \int p(x \mid u, x^-) \, bel(x^-) \, dx^- \\ &= \int p(x \mid u, x^-) \sum_i w_i \, \delta_{s^{[i]}}(x^-) \, dx^- = \sum_i p(x \mid u, s^{[i]}) w_i \end{aligned}$$

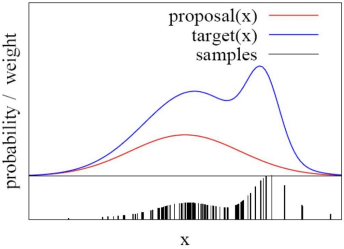
Rejection Sampling

- Let us assume that $f(x) < a$ for all x
- Sample x from a uniform distribution
- Sample c from $[0, a]$
- if $f(x) > c$ – keep the sample
- otherwise reject the sample



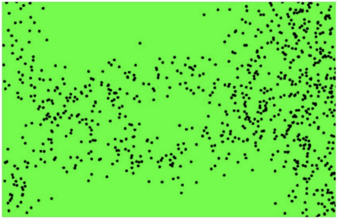
Importance Sampling Principle

- We can even use a different distribution g to generate samples from f
- By introducing an importance weight w, we can account for the "differences between g and f"

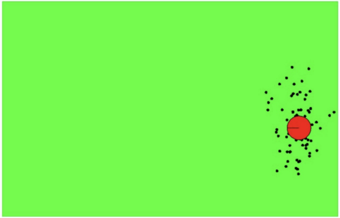


- $w = f/g$
- f is called target
- g is called proposal
- Pre−condition
  $f(x) > 0 \rightarrow g(x) > 0$

Importance Sampling with Resampling

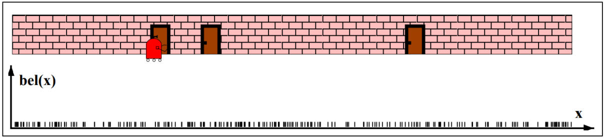


Weighted samples



After resampling

Particle Filters

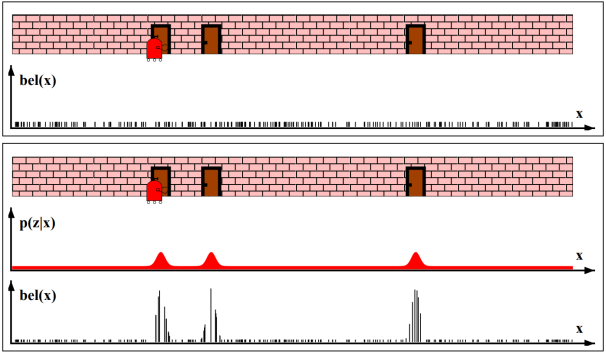


Sensor Information

Importance Sampling

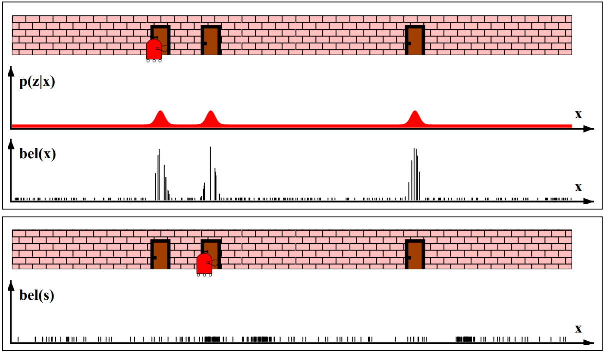
$$Bel(x) \leftarrow \eta\, p(z \mid x)\, Bel^-(x)$$

$$w \leftarrow \frac{\eta\, p(z \mid x)\, Bel^-(x)}{Bel^-(x)} = \eta\, p(z \mid x)$$



Robot Motion

$$Bel^{-}(x) \leftarrow \int p(x \mid u, x')\, Bel(x')\, dx'$$



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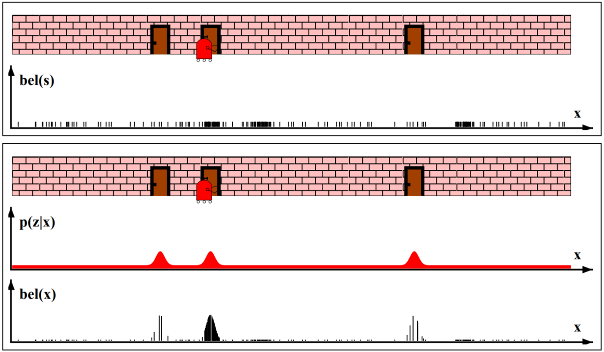

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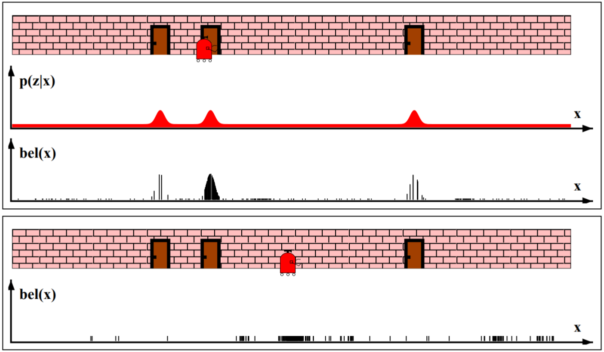


Particle Filter Algorithm

- Sample the next generation for particles using the proposal distribution
- Compute the importance weights

$$weight = target\ distribution\ /\ proposal\ distribution$$

- Resampling: “Replace unlikely samples by more likely ones”

Particle Filter Algorithm

Algorithm `particle_filter` (S_{t-1}, u_t, z_t):

$S_t = \emptyset, \qquad \eta = 0$

For $i = 1, \ldots, n$ *Generate new samples*

Sample index $j(i)$ from the discrete distr. given by w_{t-1}

Sample x_t^i from $p(x_t \mid x_{t-1}, u_t)$ using $x_{t-1}^{j(i)}$ and u_t

$w_t^i = p(z_t \mid x_t^i)$ *Compute importance weight*

$\eta = \eta + w_t^i$ *Update normalization factor*

$S_t = S_t \cup \{< x_t^i, w_t^i >\}$ *Add to new particle set*

For $i = 1, \ldots, n$

$w_t^i = w_t^i \,/\, \eta$ *Normalize weights*

Particle Filter Algorithm

$$Bel(x_t) = \eta\, p(z_t \mid x_t) \int p(x_t \mid u_t, x_{t-1})\, Bel(x_{t-1})\, dx_{t-1}$$

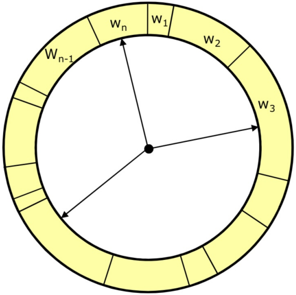
- $Bel(x_{t-1})$ – draw x_{t-1}^i from $Bel(x_{t-1})$
- $p(x_t \mid u_t, x_{t-1})$ draw x_t^i from $p(x_t \mid x_{t-1}^i, u_t)$
- $p(z_t \mid x_t)$ – importance factor for x_t^i

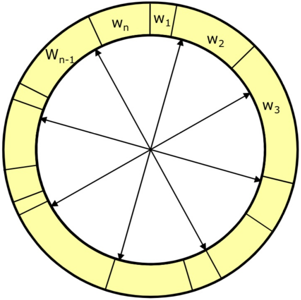
$$w_t^i = \frac{target\ distr}{proposal\ distr} = \frac{\eta\, p(z_t \mid x_t)\, p(x_t \mid x_{t-1}, u_t)\, Bel(x_{t-1})}{p(x_t \mid x_{t-1}, u_t)\, Bel(x_{t-1})} \propto p(z_t \mid x_t)$$

Resampling

- Given – Set S of weighted samples
- Wanted – Random sample, where the probability of drawing x_i is given by w_i
- Typically done n times with replacement to generate new sample set S'

Resampling





- Roulette wheel
- Binary search, $n \log n$

- Stochastic universal sampling
- Systematic resampling
- Linear time complexity
- Easy to implement, low variance

Resampling Algorithm

a.k.a. Stochastic Universal Sampling

Algorithm `systematic_resampling` (S, n):

$S' = \emptyset, \quad c_1 = w^1$

For $i = 2, \ldots, n$ — *Generate CDF*

$c_i = c_{i-1} + w^i$

$u_i \sim U[0, n^{-1}],\ i = 1$ — *Initialize threshold*

For $j = 1, \ldots, n$ — *Draw samples…*

While $u_j > c_i$ — *Skip until next threshold reached*

$i = i + 1$

$S' = S' \cup \{< x^i, n^{-1} >\}$ — *Insert*

$u_{j+1} = u_j + n^{-1}$ — *Increment threshold*

Return S'

Mobile Robot Localization

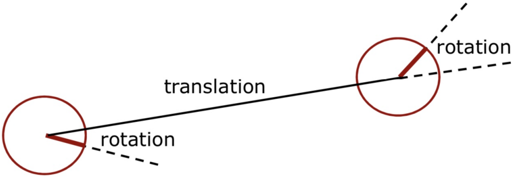
- Each particle is a potential pose of the robot
- Proposal distribution is the motion model of the robot (prediction step)
- The observation model is used to compute the importance weight (correction step)

Motion Model

Reminder

- According to the estimated motion

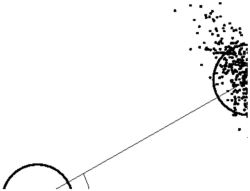
Motion Model
Reminder



- Decompose the motion into
  - Traveled distance
  - Start rotation
  - End rotation

Motion Model

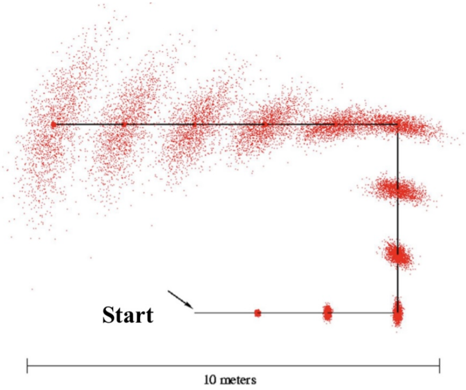
Reminder



- Uncertainty in the translation of the robot – Gaussian over the traveled distance
- Uncertainty in the rotation of the robot – Gaussian over start and end rotation
- For each particle, draw a new pose by sampling from these three individual normal distributions

Motion Model

Reminder



Mobile Robot Localization (1)
Using Particle Filters

- Each particle is a potential pose of the robot
- The set of weighted particles approximates the posterior belief about the robot's pose (target distribution)

Mobile Robot Localization (2)

Using Particle Filters

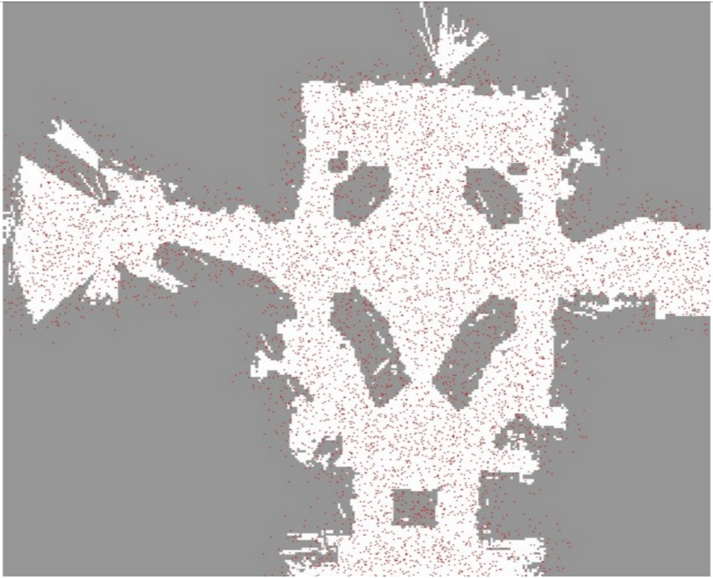
- Particles are drawn from the motion model (proposal distribution)
- Particles are weighted according to the observation model (sensor model)
- Particles are resampled according to the particle weights

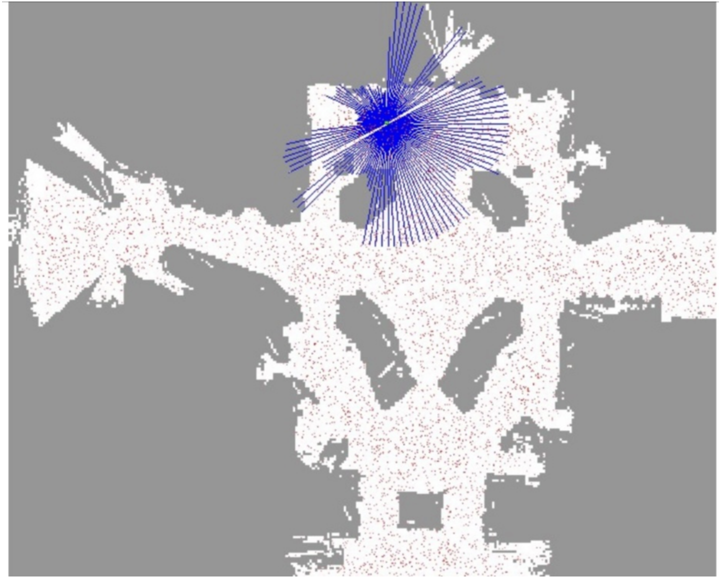
Mobile Robot Localization (3)

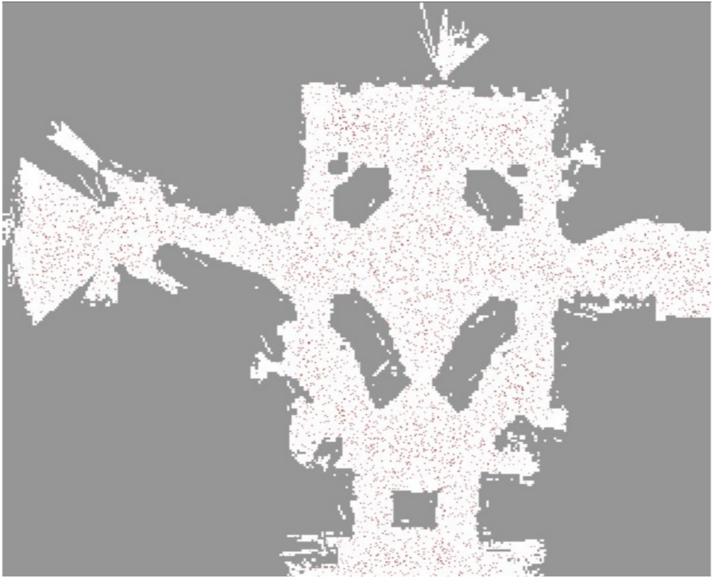
Using Particle Filters

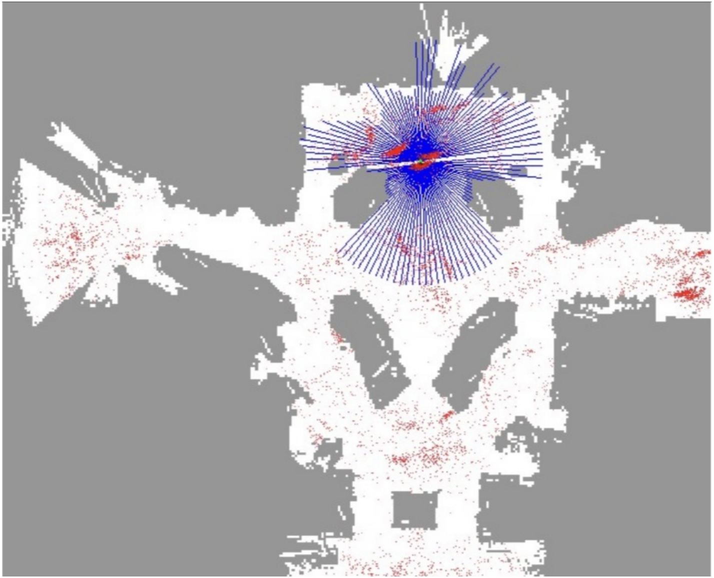
Why is resampling needed?

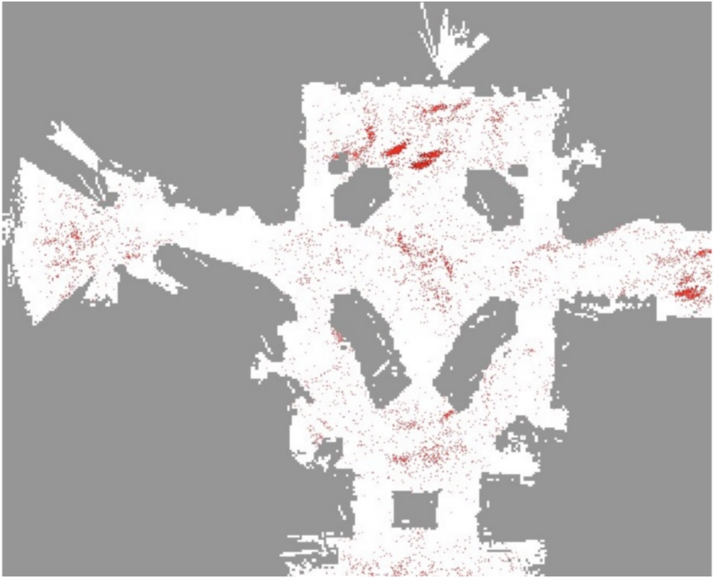
- We only have a finite number of particles
- Without resampling: The filter is likely to loose track of the "good" hypotheses
- Resampling ensures that particles stay in the meaningful area of the state space

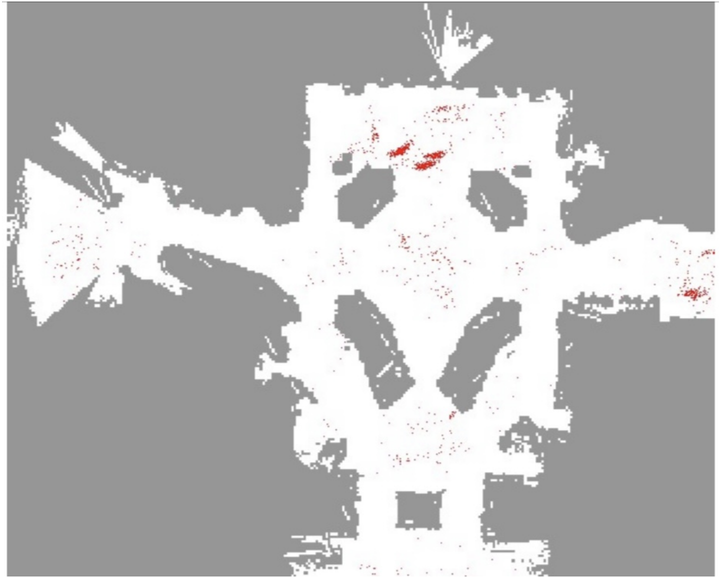


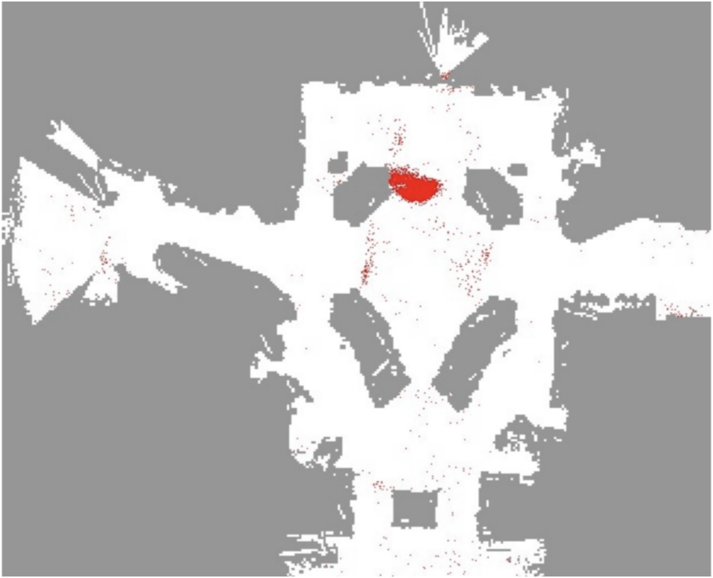


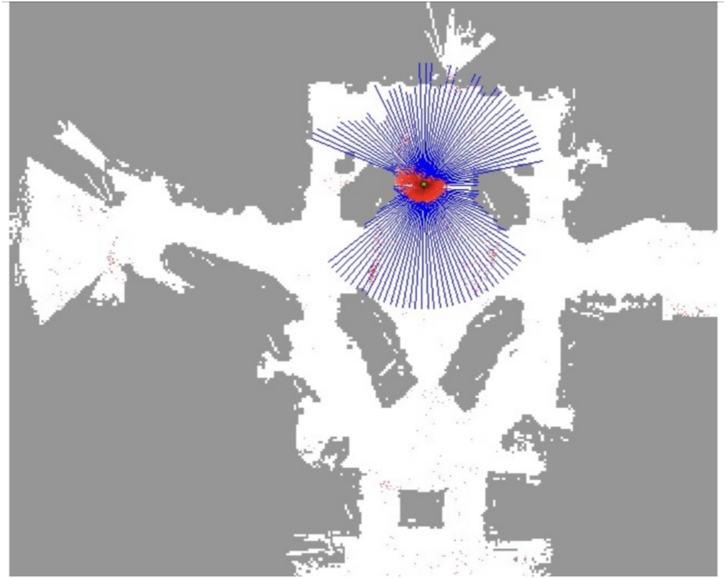


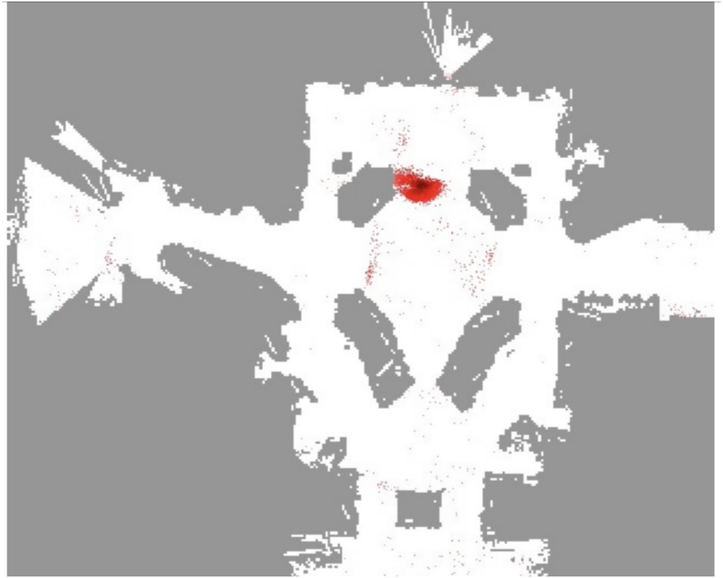


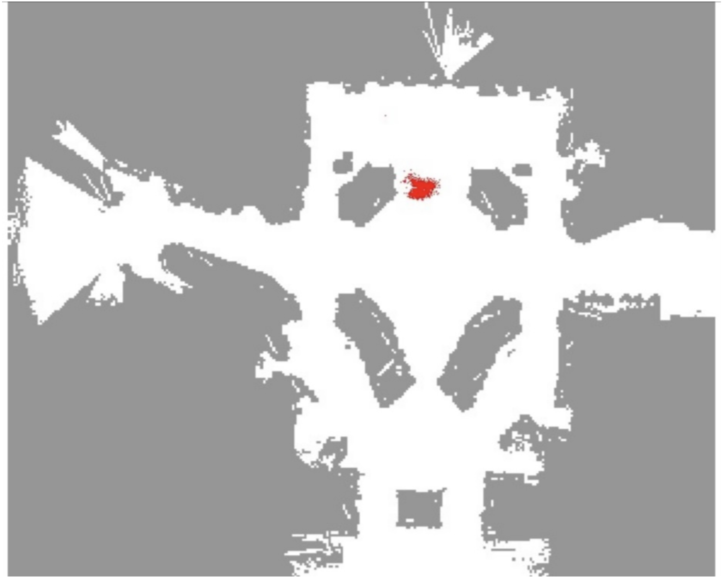


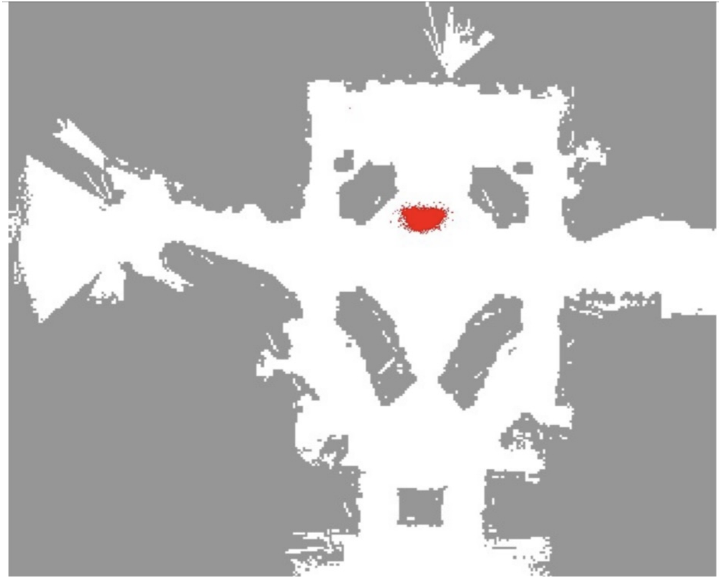


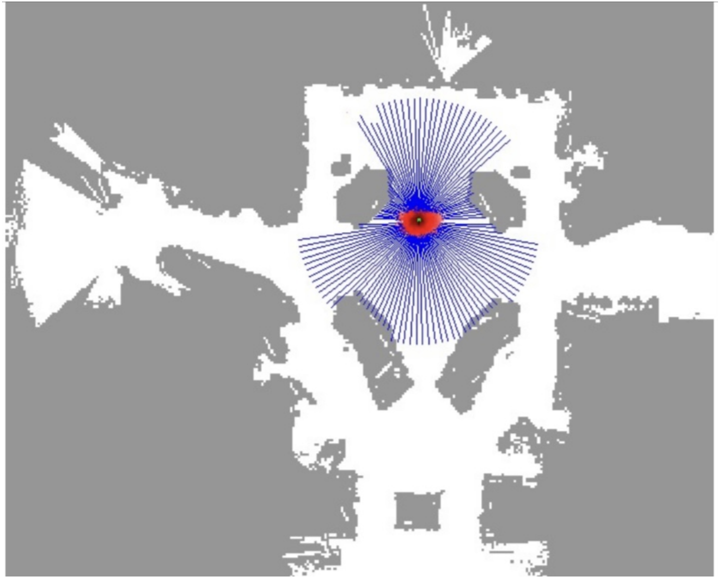


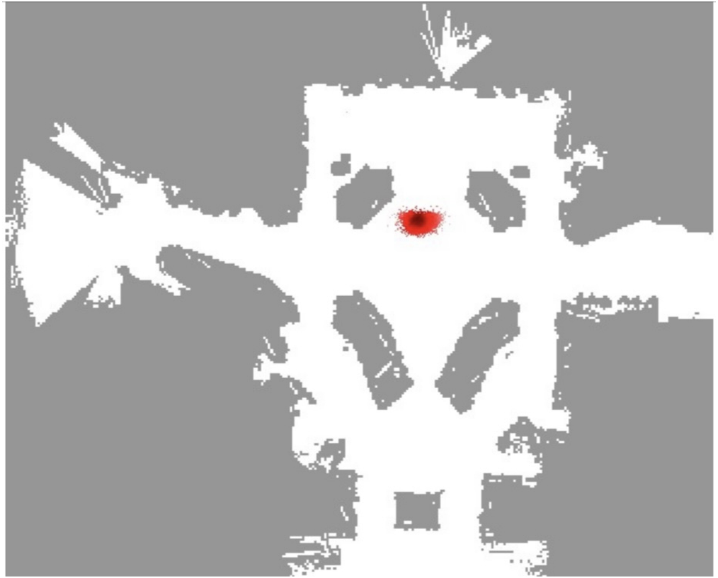


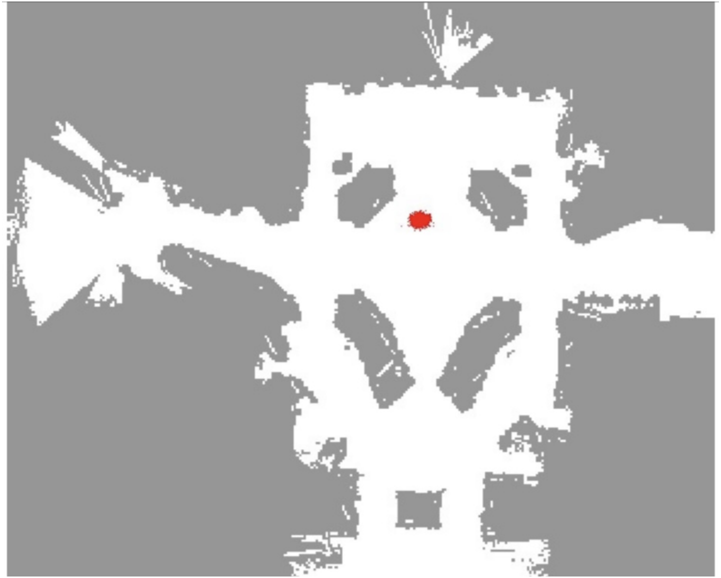


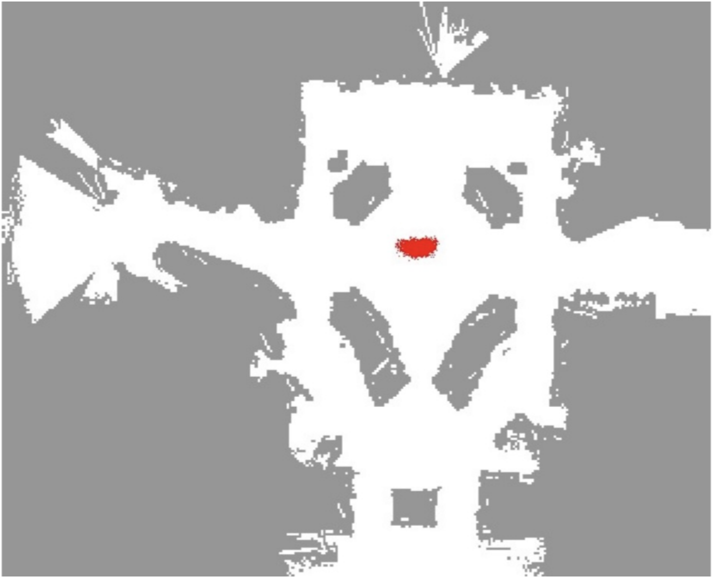


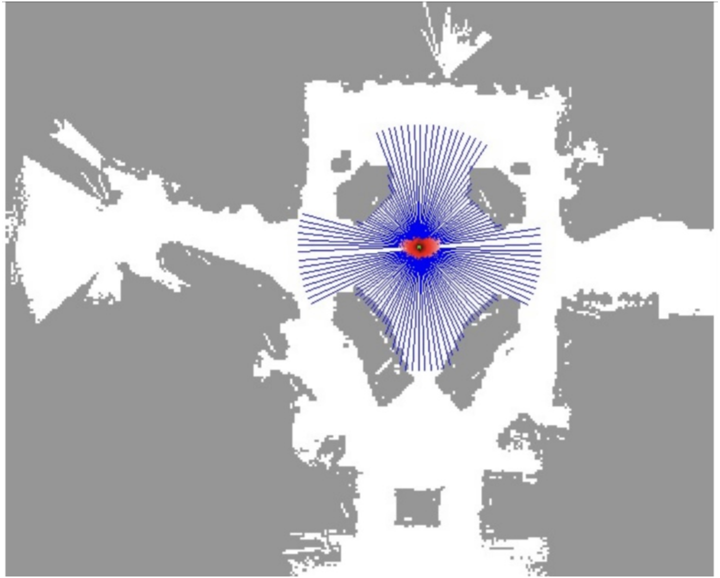






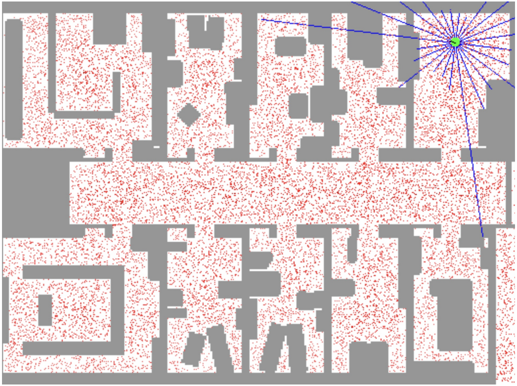




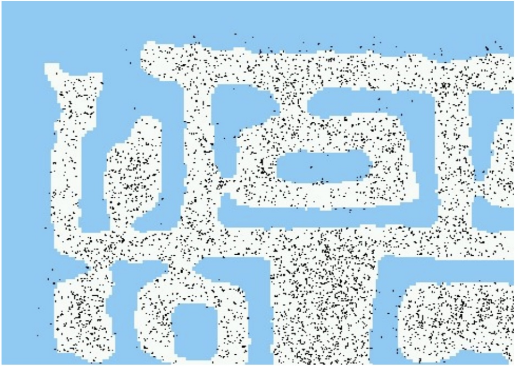


Sample–based Localization

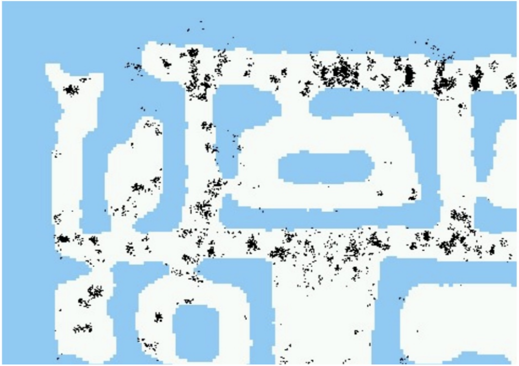
Sonar



Initial Distribution

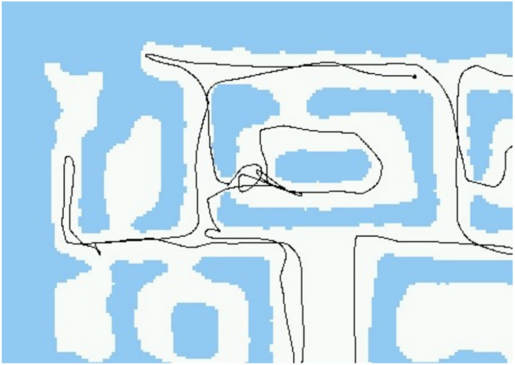


After Incorporating 10 Ultrasound Scans



After Incorporating 65 Ultrasound Scans

Estimated Path



Limitations

- The approach described so far is able
  - to track the pose of a mobile robot and
  - to globally localize the robot
- How can we deal with localization errors (i.e., the kidnapped robot problem)?

Limitations

- Randomly insert a fixed number of samples with randomly chosen poses
- This corresponds to the assumption that the robot can be teleported at any point in time to an arbitrary location
- Alternatively, insert such samples inverse proportional to the average likelihood of the observations (the lower this likelihood the higher the probability that the current estimate is wrong)

Summary

Particle Filters

- Particle filters are an implementation of recursive Bayesian filtering
- They represent the posterior by a set of weighted samples
- They can model arbitrary and thus also non−Gaussian distributions
- Proposal to draw new samples
- Weights are computed to account for the difference between the proposal and the target
- Monte Carlo filter, Survival of the fittest, Condensation, Bootstrap filter

Summary
PF Localization

- In the context of localization, the particles are propagated according to the motion model
- They are then weighted according to the likelihood model (likelihood of the observations)
- In a re−sampling step, new particles are drawn with a probability proportional to the likelihood of the observation
- This leads to one of the most popular approaches to mobile robot localization